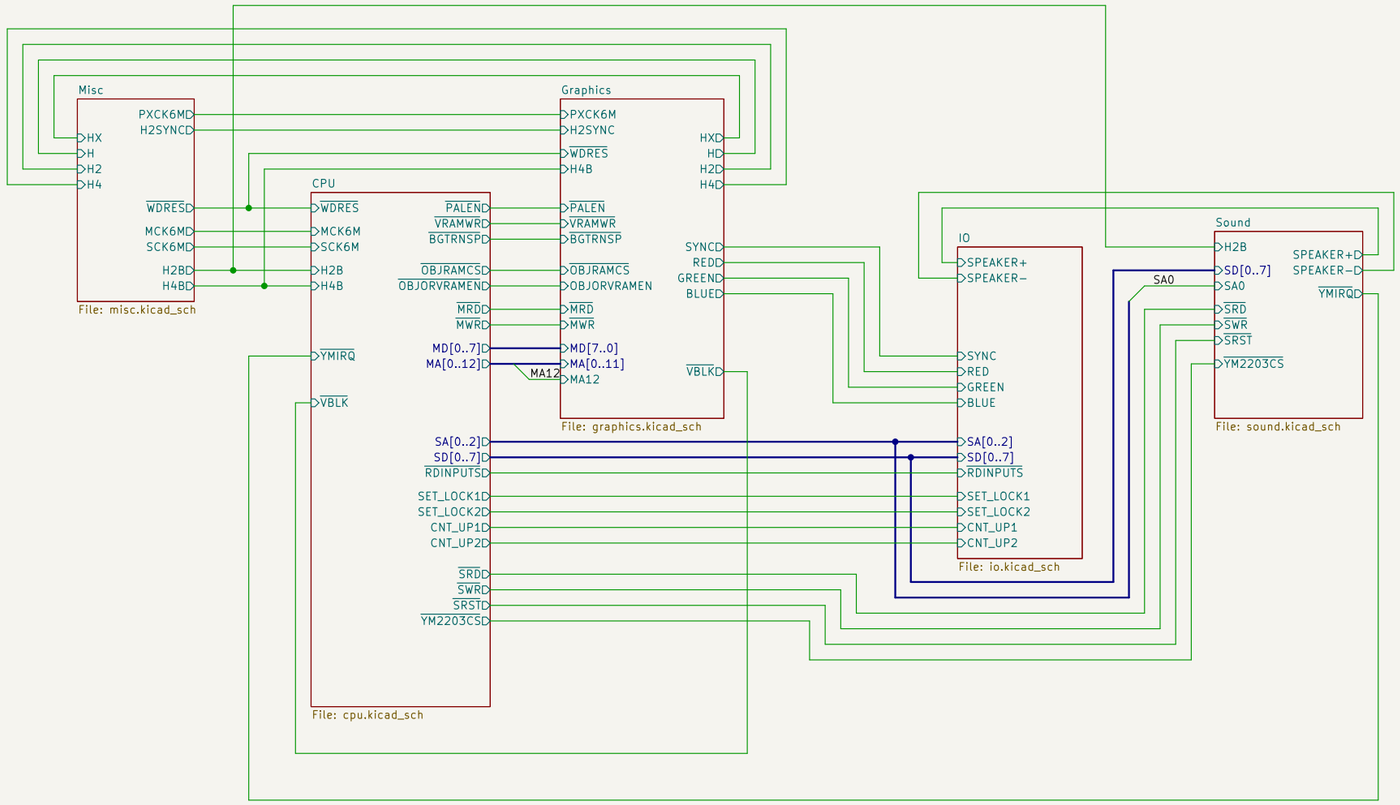


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Sheet: /
File: insectx.kicad_sch

Title: Insector X – P0–056A

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U26-B06-12

SWAIT = i9 & -i11 & (-SSHRAMEN + SSHRAMRQ)
MWAIT = OBJRVRAMEN & SSHRAMEN + i6 & SSHRAMEN
SSHRAMEN = SSHRAMRQ + -MSHRAMRQ & SSHRAMEN
MSHRAMEN = MSHRAMRQ + -SSHRAMRQ & MSHRAMEN
SHRAMCS = MSHRAMRQ & SSHRAMRQ
SHRAMWR = MWR & SSHRAMEN + MSHRAMRQ & SSHRAMRQ + SWR & MSHRAMEN
SSHRAMENB = SSHRAMEN

Inputs i7 and i8 have been set to logical 1 in the equations.

U36-B06-101

-PALEN = MA9 & MA10 & MA11 & MA12 & MA13 & -MMREQ & MRFSH
-VRAMWR = MA9 & MA10 & MA11 & MA12 & -MA13 & -MA14 & -MMREQ & MRFSH
-BGTRNSP = MA9 & MA10 & MA11 & MA12 & -MA13 & MA14 & -MA15 & -MMREQ & MRFSH & MWR
-BNKSW = MA9 & MA10 & MA11 & MA12 & -MA13 & MA14 & MA15 & -MMREQ & MRFSH & -MWR
-OBJCODELO = MA9 & MA10 & -MA11 & -MA12 & -MMREQ & MRFSH
-OBJCODEHI = MA9 & MA10 & -MA11 & MA12 & -MMREQ & MRFSH
-MSHRAMRQ = MA9 & MA10 & MA11 & -MA12 & -MMREQ & MRFSH
-OBJORVRAMEN = MA9 & MA10 & MA11 & MA12 & -MA13 & (-MA15 & -MMREQ & MRFSH + -MA14 & -MMREQ & MRFSH) +
MA9 & MA10 & -MA11 & -MMREQ & MRFSH

U35-B53-12

MRA16 = MA15 & MBSWD2
MRA15 = MA15 & MBSWD1
-MRA14 = MA15 & -MBSWD0 + -MA15 & -MA14
-MROMCS = -MA15 & MA14 & -MMREQ & MRFSH & -MBSWD2 & MBSWD1 +
-MA14 & -MMREQ & MRFSH & -MBSWD2 & MBSWD1 +
-MA15 & MA14 & -MMREQ & MRFSH & MBSWD2 +
-MA15 & -MMREQ & MRFSH & -MBSWD1 +
-MA14 & -MMREQ & MRFSH & MBSWD2
-MRAMCS = MA15 & -MA14 & -MMREQ & MRFSH & -MBSWD2 & -MBSWD1
o17 = OBJCODELO & -H4B
o18 = OBJCODEHI & -H4B
OBJRAMCS = OBJCODELO & OBJCODEHI & -H4B

U25-B06-13

-SROMCS = -SA14 & -SA13 & -SMREQ & SRFSH + -SA15 & SA14 & -SMREQ & SRFSH + -SA15 & -SA14 & -SMREQ & SRFSH
-SSHRAMRQ = SA15 & SA14 & SA13 & -SA12 & -SMREQ & SRFSH + SA15 & SA14 & -SA13 & SA12 & -SMREQ & SRFSH
-SBNKSW = SA15 & -SA14 & SA13 & -SA12 & -SWR & -SMREQ & SRFSH
-YM2203CS = SA15 & -SA14 & SA13 & SA12 & -SMREQ & SRFSH
-RDINPUTS = SA15 & SA14 & -SA13 & -SA12 & -SMREQ & SRFSH
-o17 = SA15 & SA14 & SA13 & SA12 & -SMREQ & SRFSH
SRA13 = SA15 & SBANK0 + -SA15 & SA13
SRA14 = SA15 & SBANK1 + -SA15 & SA14

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Sheet: /CPU/	
File: cpu.kicad_sch	
Title: Insector X – P0–056A	
Size: A3	Date: 2022–12–06
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	Rev:

U40
74LS245
U41
74LS374
U29D
74LS32
U11A
74LS04
U42
TMM2063AP_10
U11D
74LS04
U29B
74LS32
U11E
74LS04
74LS157
U16
U23
74LS245
U33
74LS245
U10
74LS374
U9
74LS174
OSC1
OSC-12.0000MHz
R1
330
R2
22
U24-1
YM3906
SETA_X1-001
U24-2
YM3909
SETA_X1-002
U50
X1-006
U45
X1-007
U48C
74LS14
U54
74LS161
U2
LH53400
U1
LH53400
MD[7..0]
VD[0..7]
VDB[0..7]
VA[0..11]
MA[0..11]
MA[0..9]
GA[0..4]
GA[5..17]
G2D[0..15]
GD[0..15]
OBJORVRAMEND
H4BD
MRDD
H4
H2SYNCD
OBJRD
OBJRAMCSD
VRAMWRD
WDRESD
H8
HBLK
HSYNC
VSYNC
SYNC
PXCK6MD
VBLK
RED
GREEN
BLUE
R
G
B
R0-012
SYNC
C18
330pF
R16
47
Pixel clock PXCK6M: f = 6MHz

Horizontal sync and blanking signals have a
period of 64us and frequency 15.625kHz.
Pixels / horizontal line = 64us x 6MHz = 384 pixels.

Vertical sync and blanking signals have a
period of 16.896ms and a frequency of 59.1856Hz.
Vertical lines = 16.896ms x 15.625kHz = 264 lines.

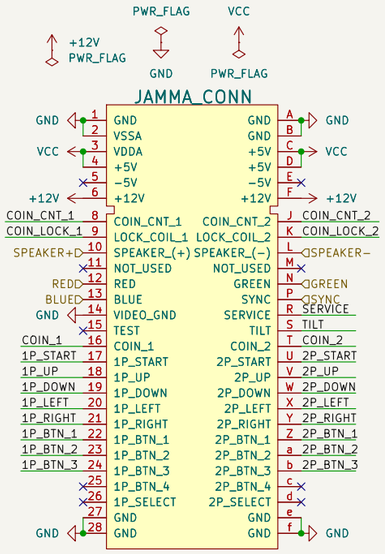
The composite SYNC signal is a wired and of HSYNC and VSYNC

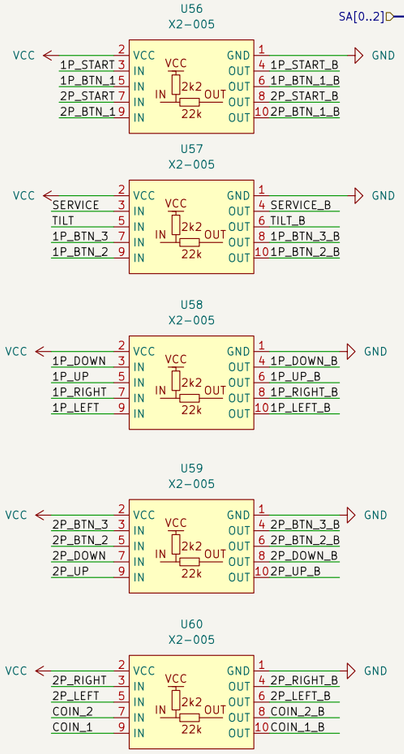
	HBLK	HBLK2	HSYNC
Active pixels	320	256	352
Blank pixels	64	128	32

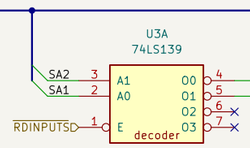
	VBLK	VSYNC
Active lines	224	256
Blank lines	40	8

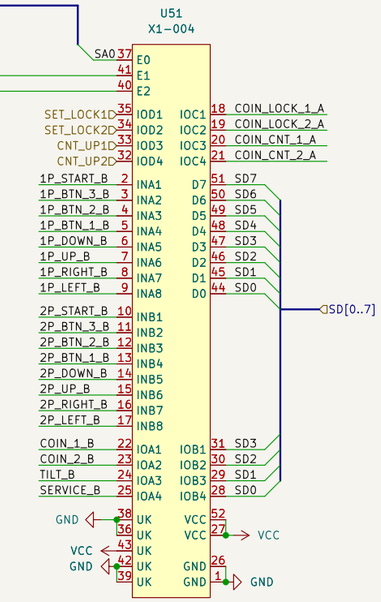
	HBLK2-HBLK	HBLK-HSYNC	VBLK-VSYNC
Front porch	32	4	16
Back porch	32	28	24

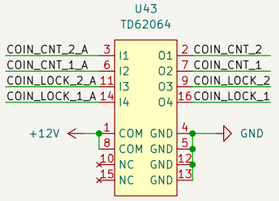
When VSYNC goes low there is a delay of 60 pixels (10us x 6MHz)
from HSYNC going high. It is the same delay when VSYNC goes high.
U23 and U33 use a modified version of LS245
in order to avoid errors during ERC. This happens
when GND is connected to a tri-state input.
10us = 60 pixels
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Sheet: /Graphics/
File: graphics.kicad_sch
Title: Insector X - P0-056A
Size: A3
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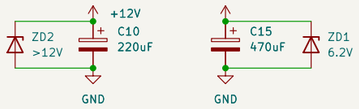


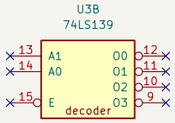
The I/O direction of each port in X1-004 is probably determined by connecting the pins 36,38,39, 42 and 43 to VCC and GND accordingly.

In insectx IOD -> IOC, IOA -> IOB

E2 E1 E0 Active output
H L L INA-> D, IOD -> IOC
H L H INB -> D, IOD -> IOC
L H L IOA -> IOB
X X X IOD -> IOC

All inputs are active low, when the outputs are not active they are in tri-state state. Except for IOC in this case.



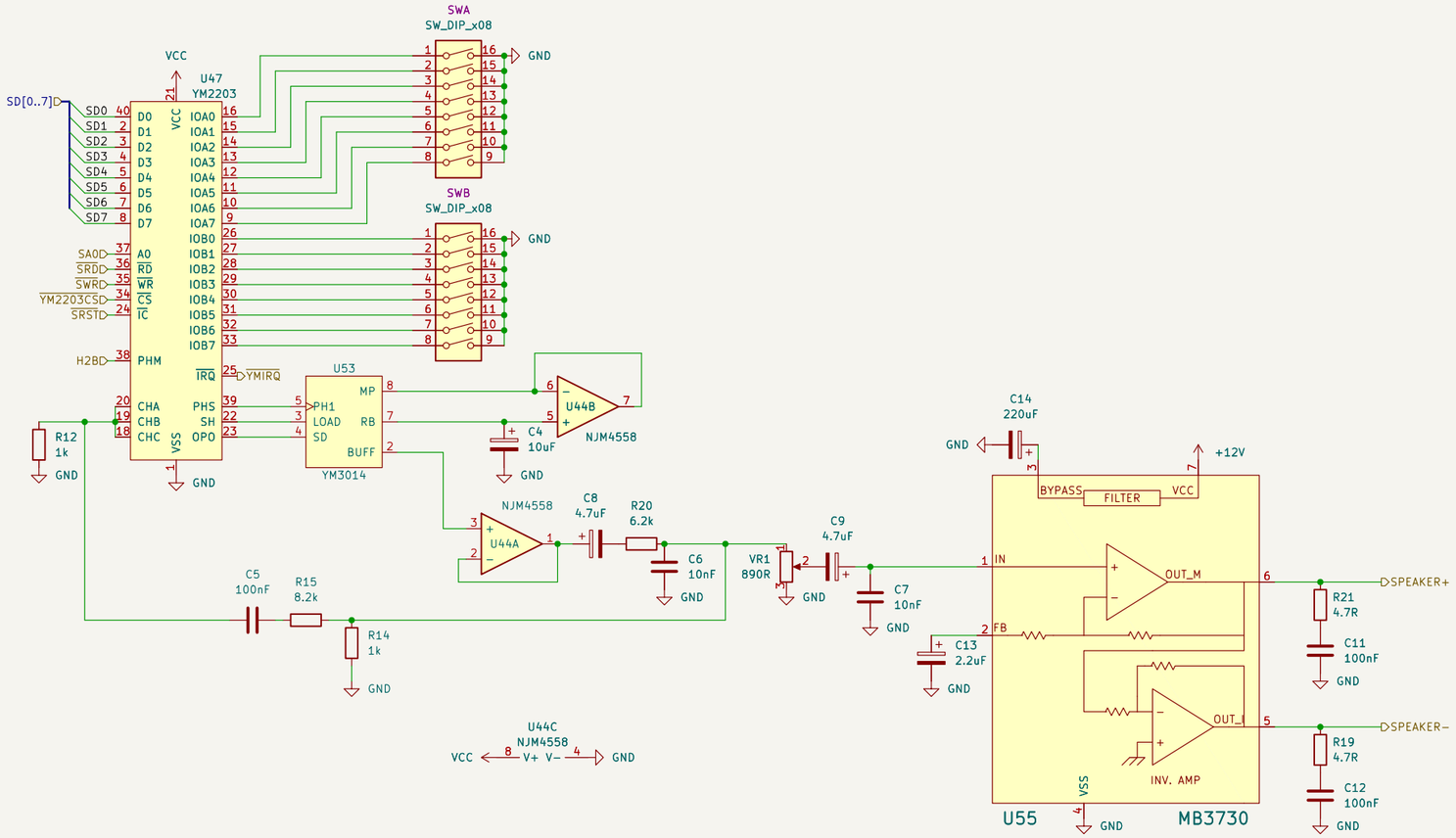


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Sheet: /IO/
File: io.kicad_sch

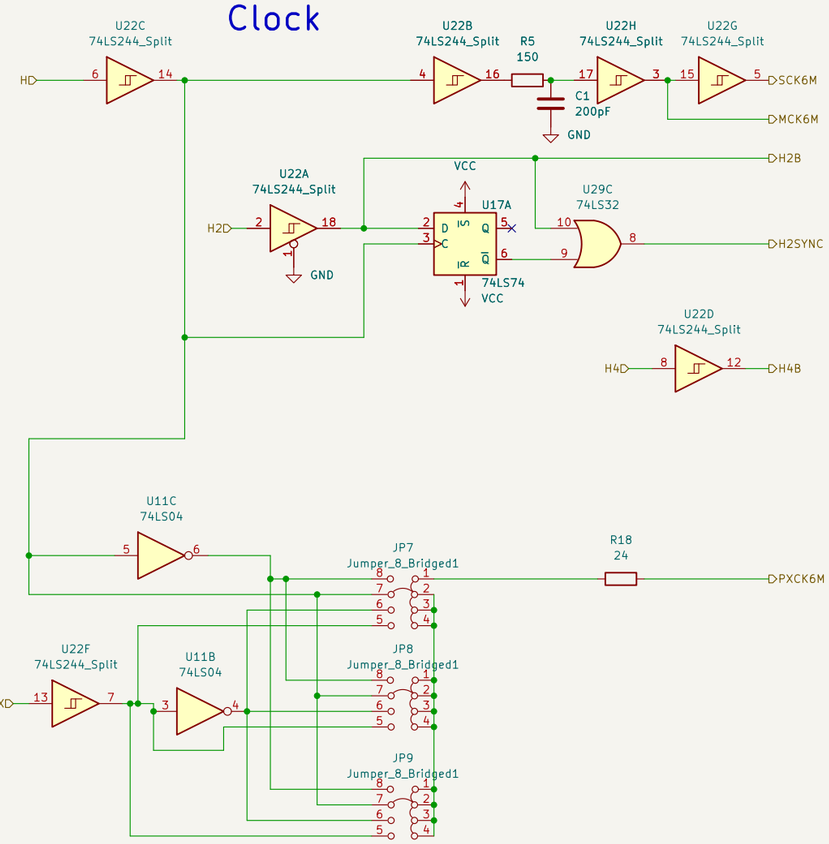
Title: Insector X – P0-056A

Size: A3	Date: 2022-12-06	Rev:
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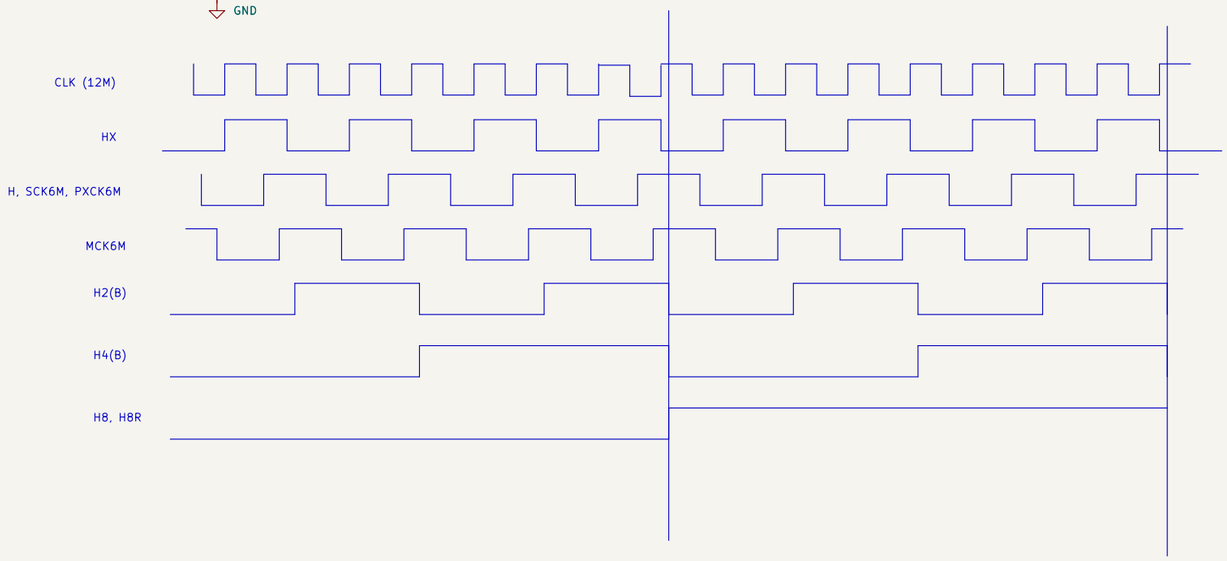


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Sheet: /Sound/		
File: sound.kicad_sch		
Title: Insector X – P0–056A		
Size: A3	Date: 2022–12–06	**Rev:**
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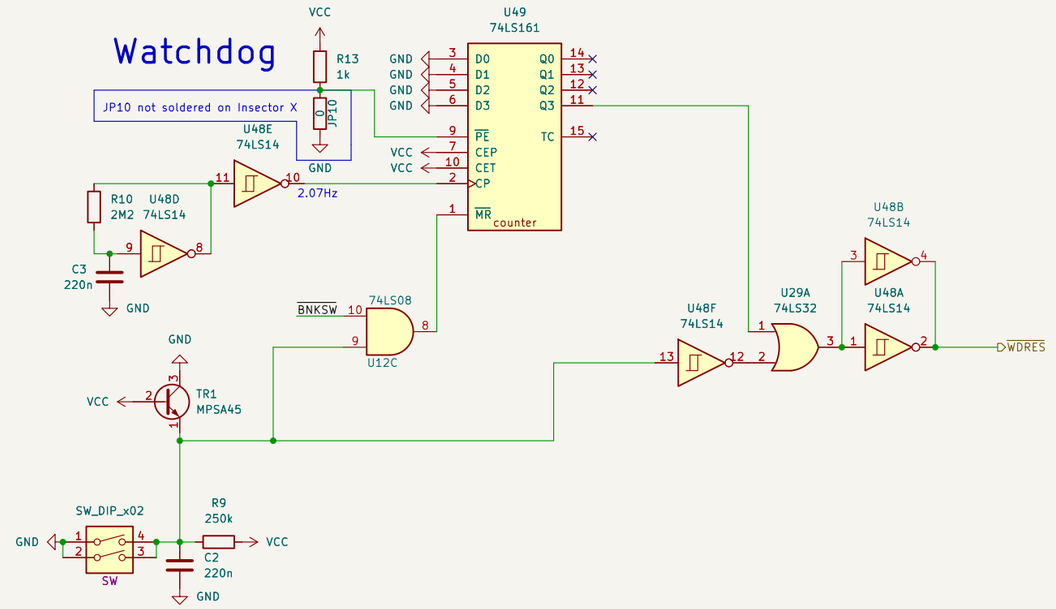
Clock







Watchdog



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Sheet: /Misc/ File: misc.kicad_sch		
Title: Insector X – P0–056A		
Size: A3	Date: 2022–12–06	**Rev:**
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